



ALL



1. compscience224

Which of the following statements is false?

Pick **ONE** option

1

☐ Encryption is a reversible operation

2

☐ The amount of change in the output hash is proportional to the change in the input data

3

☐ Hash is an irreversible operation

4

☐ different encrypted strings can result in the same cipher text but should not result in the same not in the same hash text for a perfect hashing algorithm

5

Clear Selection

6

7

8

- ☐ different encrypted strings can result in the same cipher text but should not result in the same not in the same hash text for a perfect hashing algorithm

Clear Selection

2. Aptitude004

"Pedro goes either hunting or fishing every day. If it is snowing and windy then Pedro goes hunting. If it is sunny and not windy then Pedro goes fishing. Sometimes it can be snowing and sunny. Which of the following statements must be true?"

Pick **ONE** option

☐ If it is not sunny and it is snowing, then Pedro goes hunting.

☐ If it is windy and Pedro does not go hunting then it is not snowing.

☐ If it is windy and not sunny then Pedro goes hunting.

☐ If it is windy and sunny then Pedro goes hunting.

Clear Selection

3. compscience222

☐ If it is windy and sunny then Pedro goes hunting.

Clear Selection

3. compscience222

Which of these is the most secure way of authenticating an API?

Pick **ONE** option

☐ A unique API key per application

☐ HTTP Basic Authentication

☐ OAuth2 token

☐ NT Lan Manager authentication

Clear Selection

4. Aptitude027

5. A clock runs not accurately at 10am. When the clock indicates 7:30pm, the true time is:

Clear Selection

4. Aptitude027

A watch gains 6 seconds every 5 minutes was set accurately at 10am. When the clock indicates 7:30pm, the true time is:

Pick **ONE** option

☐ 7:00 PM

☐ 7:10 PM

☐ 7:(960/51) pm

☐ 7:(960/48) pm

Clear Selection

5. Networking082

Detection mechanism?

☐ 7:(960/48) pm

Clear Selection

5. Networking082

Which of the following is not an error detection mechanism?

Pick **ONE** option

☐ Parity check

☐ Padding

☐ Hamming code

☐ CRC

Clear Selection

6. Aptitude031

Two trains are moving towards each other. Train A travels constantly at 25km/h and B travels at 35Km/h. If the total time traveled by them is 8hr30m before they cross, what is the distance between X & Y

☐ CRC

Clear Selection

6. Aptitude031

Train A starts from X and train B starts from Y 30 minutes later towards each other. Train A travels constantly at 25km/h and B travels at 35Km/h. If the total time traveled by them is 8hr30m before they cross, what is the distance between X & Y

Pick **ONE** option

☐ 178.5

☐ 272

☐ 252.5

☐ 306.5

Clear Selection

☐ 306.5

Clear Selection

7. compscience217

Comparing HTTP based APIs and CRUD operations, which among the following statements correctly map the HTTP method with the CRUD operation:

Pick **ONE** option

☐ GET = Retrieve, POST = Create, HEAD = Delete, PUT = Update

☐ GET = Delete, POST = Update, HEAD = Create, PUT = Retrieve

☐ GET = Retrieve, POST = Create, HEAD = Update, PUT = Delete

☐ GET = Retrieve, POST = Create, DELETE = Delete, PUT = Update

Clear Selection

8. Networking059

8. Networking059

Which of these statements is true concerning fragmentation and segmentation?

Pick **ONE** option

☐ Fragmentation occurs at the network layer while segmentation at the transport layer.

☐ Fragmentation occurs at the transport layer while segmentation at the network layer.

☐ Applications are aware of fragmentation but not of segmentation.

☐ None of the above.

Clear Selection

9. compscience218

Which series will the HTTP response code belong to in case the server wants to indicate a success?

Pick **ONE** option

☐ 2xx

9. compscience218

Which series will the HTTP response code belong to in case the server wants to indicate a success?

Pick **ONE** option

☐ 2xx

☐ 3xx

☐ 4xx

☐ 5xx

Clear Selection

10. compscience219

Which series will the HTTP response code belong to in case the server wants to indicate that the client needs to take further action such as go to a new location?

Pick **ONE** option

10. compscience219

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Pick **ONE** option

☐ 2xx

☐ 3xx

☐ 4xx

☐ 5xx

Clear Selection

11. compscience225

Choose the Incorrect statement relating to public key cryptography:

☐ 4xx

☐ 5xx

Clear Selection

11. compscience225

Choose the incorrect statement relating to public key cryptography:

Pick **ONE** option

☐ Private key is used for cryptographically signing content to detect tampering

☐ Receiver's public key is used for encrypting the message which can be decrypted using the receiver's private key

☐ Sender's private key is used to encrypt the message which can be decrypted using the receiver's private key

☐ Public key cryptography is an example of asymmetric encryption

Clear Selection

12. Networking080

Which address class has no host addresses?

Pick **ONE** option

☐ Class D

☐ Class B

☐ Class C

☐ Class A

Clear Selection

13. compscience193

What does the following function do?

13. compscience193

What does the following function do?

```
Int func(int x)
{
    int val;
    val= x && !(x & (x-1));
    return (val);
}
```

Pick **ONE** option

☐ Returns 1 if x is an even number, else returns 0.

☐ Returns 1 if x is a power of 2, else returns 0.

☐ Returns 1 if x is a perfect square, else returns 0

☐ Returns 1 if x is the square-root of an even number, else returns 0.

Clear Selection

Clear Selection

14. Aptitude034

A invests 24000 at the start of the year, B invests 12000 at 6 months and C invests 6000 at 9 months in a business. If the profit at the end of the year is 21000 what's the share for B?

Pick **ONE** option

☐ 6000

☐ 2000

☐ 4000

☐ 8000

Clear Selection

15. Aptitude029

Clear Selection

15. Aptitude029

Find the next numbers in the sequence 5,45,40,10,38,28,15,?,?

Pick **ONE** option

☐ 35,20

☐ 20,16

☐ 31,16

☐ 31,20

Clear Selection

Continue

3m left

17. Balance the Brackets

A bracket is considered to be any one of the following characters: (,), {, }, [, or]. Now two brackets are considered to be a matched pair if the an opening bracket (i.e., (, [, or {) occurs to the left of a closing bracket (i.e.,),], or }) of the exact same type. There are three types of matched pairs of brackets: [], {}, and ().

A matching pair of brackets is not balanced if the set of brackets it encloses are not matched. For example, `{[([])}` is not balanced because the contents in between { and } are not balanced.

Given a string of brackets, determine whether each sequence of brackets is balanced & how many swaps are needed to make it balanced. If a string is balanced, return 0. If the string cannot be balanced return -1. Else return the number of swaps needed to balance the string.

Input format:

A string of brackets of length n.

Output format:

An integer. If a string is balanced, return 0. If the string cannot be balanced return -1. Else return the number of swaps needed to balance the string.

Constraints:

$0 < n \leq 1000$

Sample Input 0:

`{[]}`

Sample Output 0:

0

Sample Explanation 0:

This is already balanced and hence answer would be 0.

Google Chrome
il - Prateek Somani - O x HackerRank
hackerrank.com/test/eoiq3k9c95r/questions/7o8nm4pfied
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Constraints:

$0 < n \leq 1000$

Sample Input 0:

{}

Sample Output 0:

0

Sample Explanation 0:

This is already balanced and hence answer would be 0.

Sample Input 1:

{}

Sample Output 1:

1

Sample Explanation 1:

The string is not balanced. And one swap will give us {}. Answer is 1

Sample Input 2:

{ } is not balanced

Sample Output 2:

-1

Sample Explanation 2:

This string cannot be balanced with any number of swaps. Answer is -1.

17. Rumble Raffle

Jerry is playing the following game wherein the objective is to buy the full house ticket with the tokens in hand at the lowest time possible.

Tokens help player buy the full house ticket.

In this game, player starts with 0 tokens and the lender lends you 2 tokens/second.

Whenever player has X tokens, player can buy a token generator from the lender.

Every token generator gives player an extra Y tokens/second.

Once player has Z tokens which is the cost to buy the full house ticket, game is won.

Figure out the best strategy (least amount of time) to win the game.

Input

On first three lines the input would be X, Y, Z respectively.

Output

Return the lowest time it takes to buy the raffle.

Constraints

$1 \leq X \leq 1000000$

$1 \leq Y \leq 1000$

$1 \leq Z \leq 1000000$

Note: You are getting the token only at 2 tokens/second.

Sample

Input :

900 4 2000

Output :

type here to search

16. A Lot Can Happen Over Coffee

A Coffee Shop is famous for freshly brewed coffee with different customisation it provides to customers. There are many binary ("either-or") options for customising the coffee order, such as "with sugar"/"no sugar", "with ice"/"no ice" and "vanilla"/"no vanilla". A customer's preferences for their coffee can be represented as a binary string. For example, using the three properties above (in the order they are given), the string 110 means "with sugar, with ice, not vanilla".

Vikas is hosting a bunch of friends (N friends) in the coffee shop that offers P binary options. But after collecting everyone's preferences, Vikas found that the order was getting too complicated, so he decided to buy the same type of coffee for everyone. Vikas knows that for every friend, for every preference that is not satisfied, they will complain once. For example, if two of the friends have preferences for types 101 and 010, and Vikas chooses type 001, then the first friend will complain once and the second friend will complain twice, for a total of three complaints.

Moreover, there are M different forbidden types of milk coffee that the shop will not make, and Vikas cannot choose any of those forbidden types.

What is the smallest number of complaints that Vikas can get?

Input:

First line contains P which represents binary options.

Next line contains N which represents the number of friends.

Each of the next N lines contains a binary string which represent the preference of each friend.

Next line contains M which represents the number of forbidden combinations.

Each of the next M lines contains a binary string which represent the forbidden types of milk coffee that the shop will not make.

Output:

An integer y which is the minimum number of complaints that Vikas can get, per the rules described above.

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